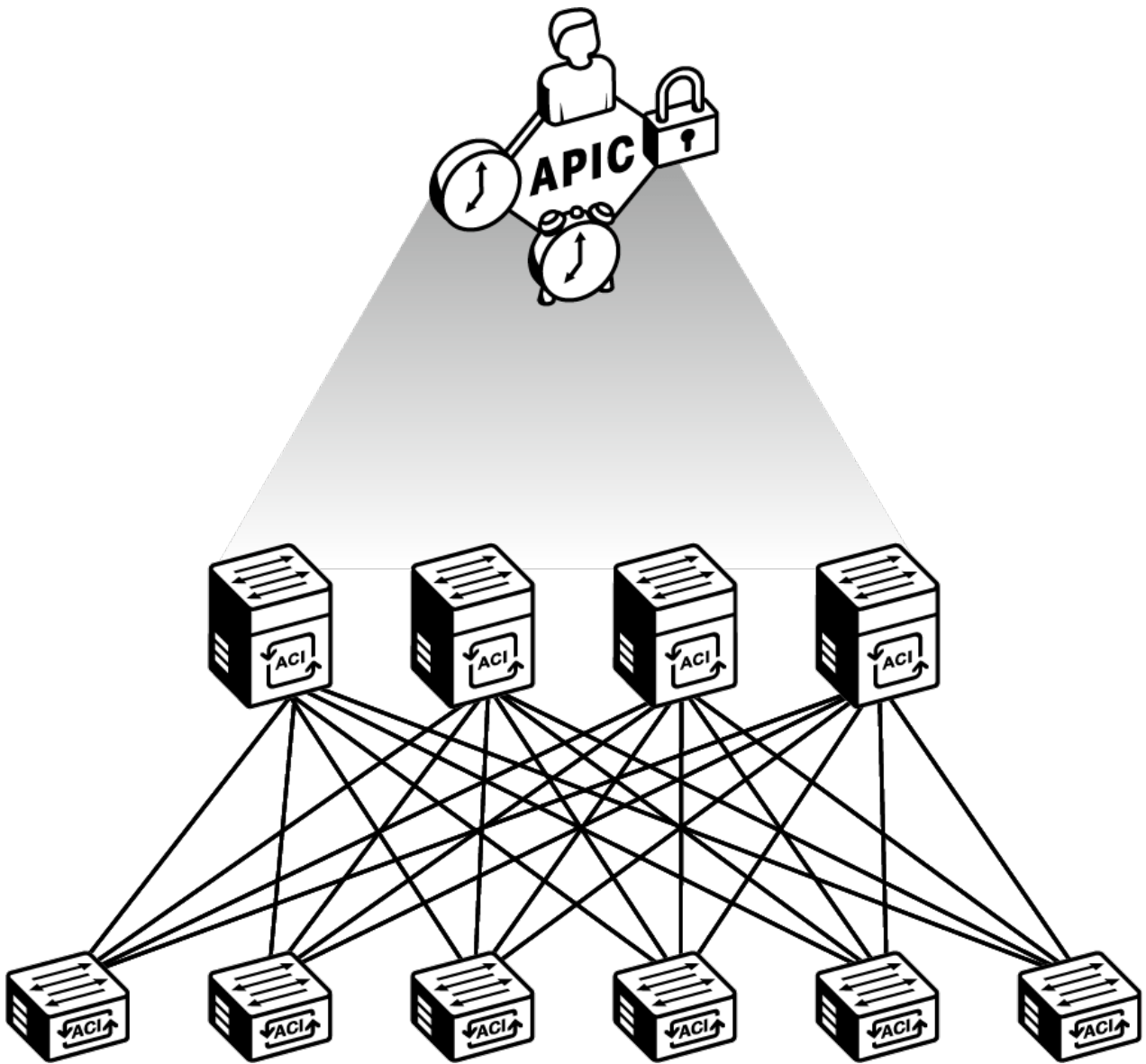


Describing Cisco ACI

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For more labs visit my GitHub repo: <https://github.com/TitusM/Cisco-Data-Center>



Note

This lab was conducted in a controlled environment. Any configurations in a production network should be implemented during a designated maintenance window. Additionally, always refer to official documentation relevant to your specific hardware and software.

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Cisco ACI Overview

Cisco ACI is a software-defined network architecture for modern data centers.

Cisco ACI is built on the following:

- Cisco Application Policy Infrastructure Controller (Cisco APIC) – orchestration and management
- Cisco ACI fabric – leafs and spines (switches for packet forwarding)

The main components of Cisco ACI are the APICs, leafs and spines.

APIC

- Manages the fabric
- Policy enforcement
- Health monitoring
- Supports API to enable programmability and automation

Leafs

- Leafs provide the connection points for endpoints, external routers, L4-L7 devices (firewalls/load balancers)
- Leafs connect to spines and not to each other

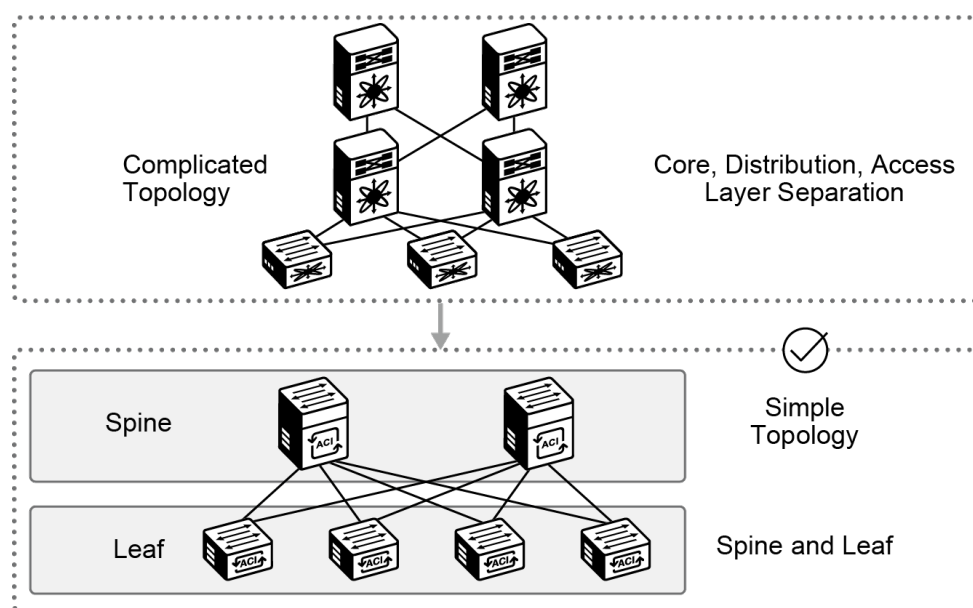
Spines

- Spines contain the databases (known as COOP - (Council of Oracle Protocol)) that enables traffic forwarding in the ACI fabric.
- Endpoints do not connect to spines
- Spines provide connection points for Data center interconnects (IPN/ISN)

Underlay routing protocol

- IS-IS

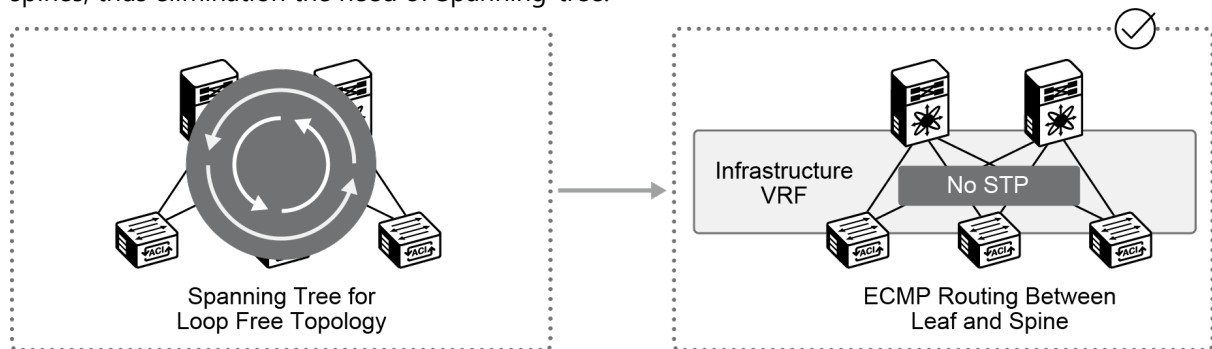
Legacy Data Center vs Cisco ACI



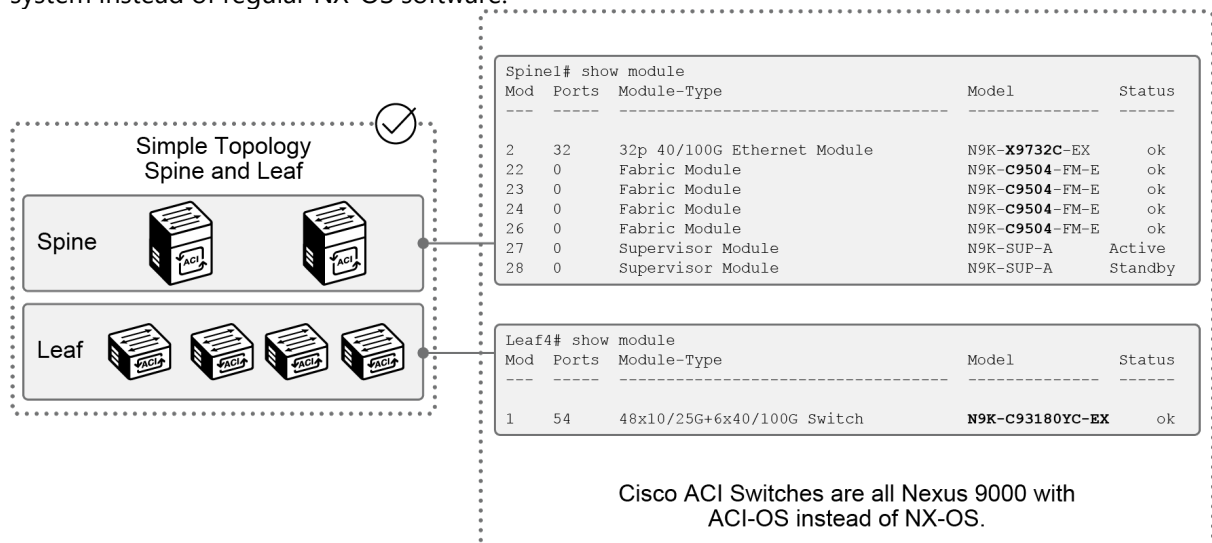
Traditional networks are usually made up of a 3-tier setup (access-aggregation-core).

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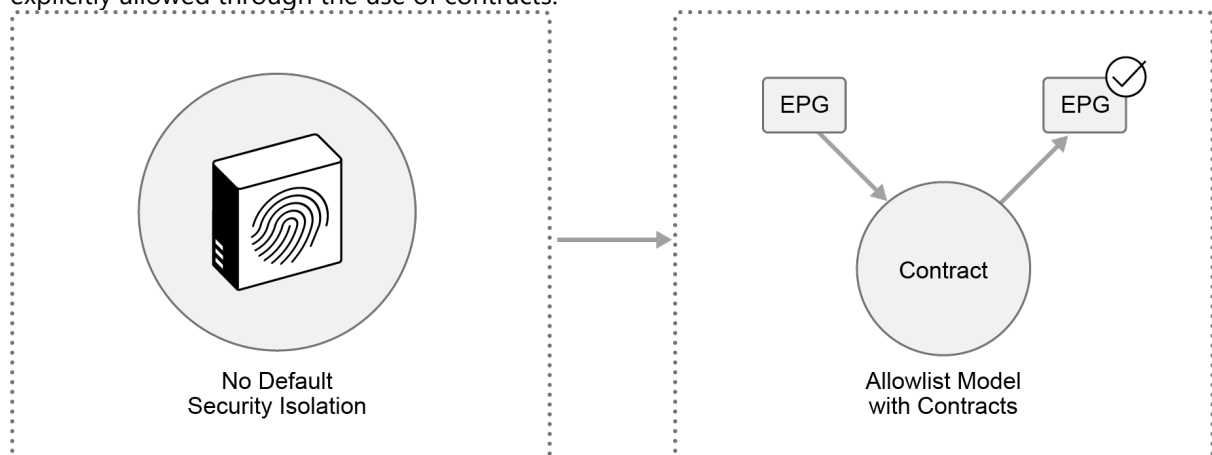
Cisco ACI uses a leaf-spine topology with a routed underlay. There is no layer 2 between leafs and spines, thus eliminating the need of Spanning-tree.



The Cisco ACI fabric is made up of leafs and spines (Nexus 9000 series) that run the ACI operating system instead of regular NX-OS software.

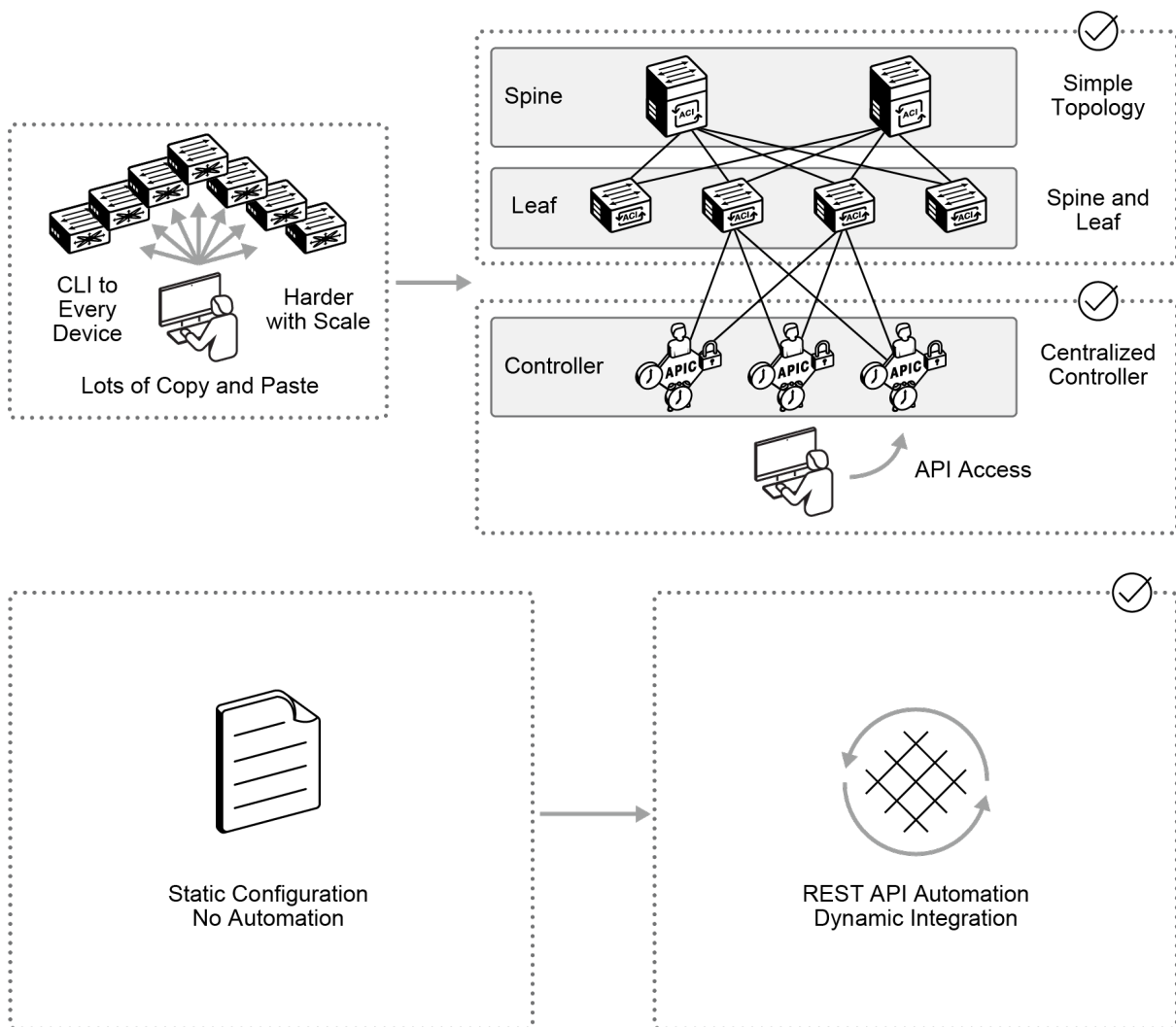


Cisco ACI employs a "whitelist model", i.e. traffic communication is blocked by default unless it is explicitly allowed through the use of contracts.



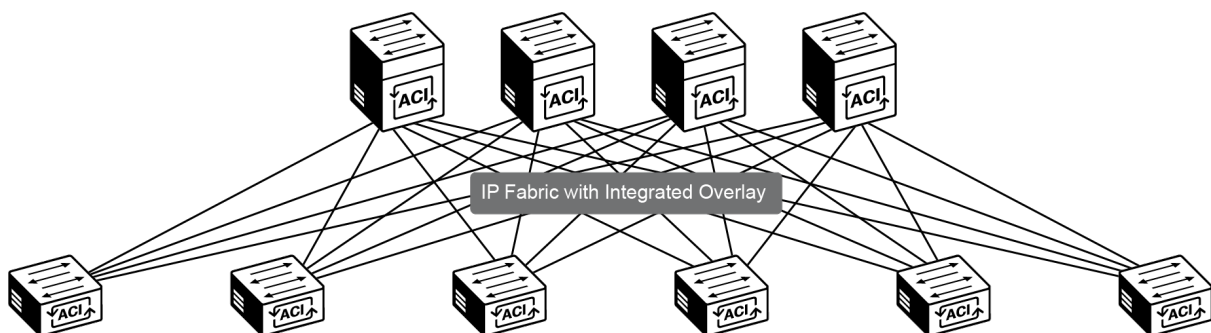
Using a controller (APIC), centralized management and configurations of devices is possible. Unlike the traditional methods of accessing each device via CLI to push configurations, the Cisco APIC is enriched with API capabilities allowing the ease access of policies and objects that represent configurations. The controller does not get involved in the data plane, thus losing the controller does not impact traffic forwarding in the fabric.

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Cisco ACI Topology & Hardware

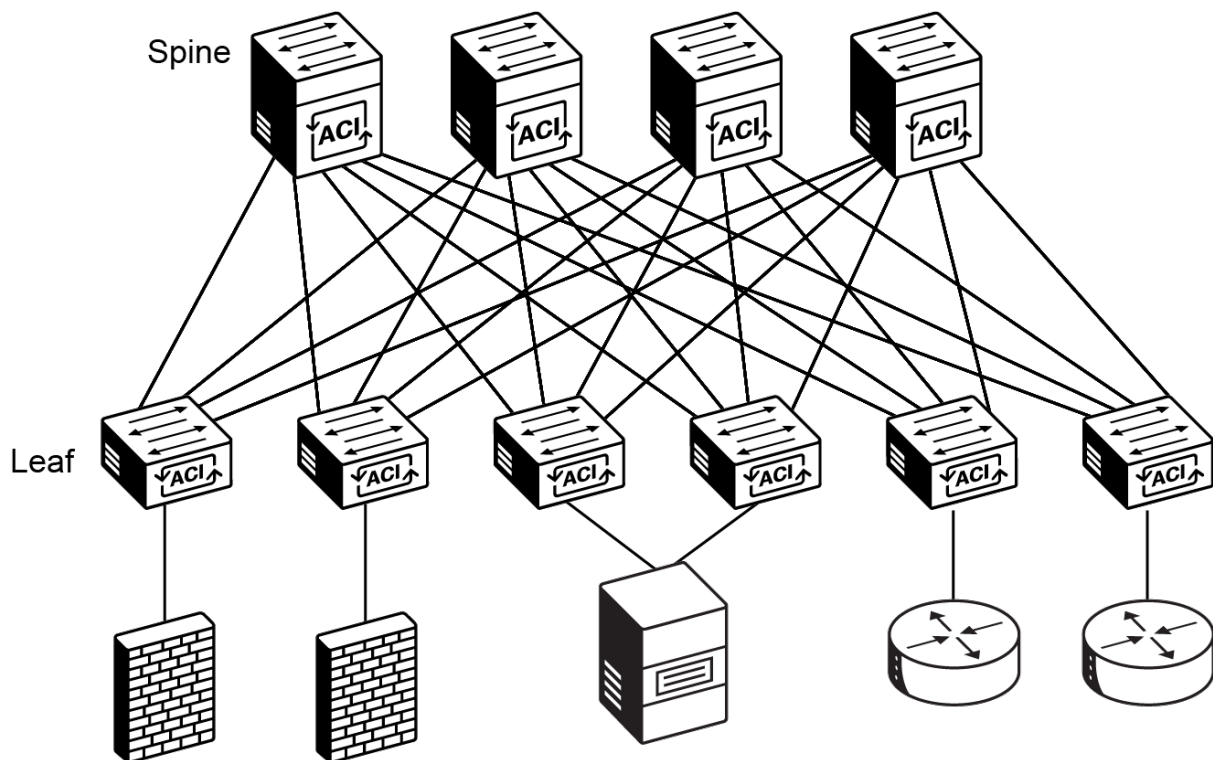
The Cisco ACI fabric is made up of a spine-leaf topology.



The leafs provide the access ports for endpoint connections whilst the spines aggregate the fabric bandwidth.

The Cisco ACI fabric is comprised of the Cisco APIC and the Cisco Nexus 9000 Series spine and leaf switches. The leaf switches connect to the spine switches but never to each other. The same goes for the spines as well. The Cisco APIC and all other endpoints and devices in the data center connect to the leaf switches only as shown in the following figure.

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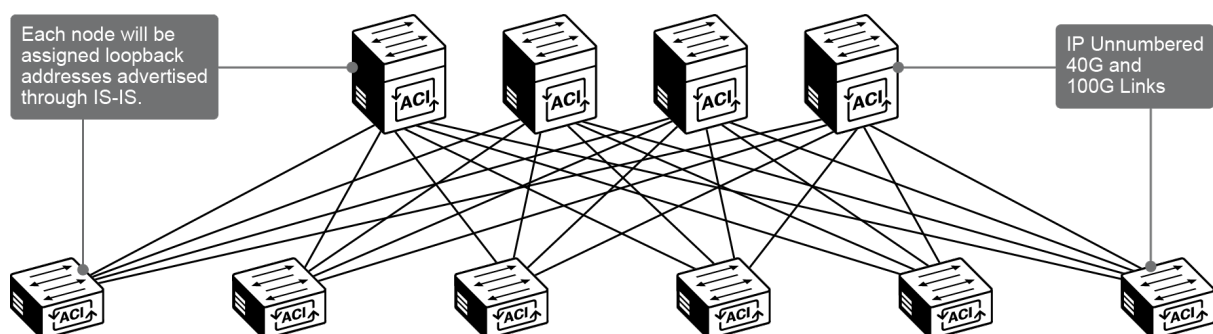


The APICs manage the infrastructure IP addressing scheme in a dedicated vrf (**infra**) and they act as DHCP servers, responsible for allocating IP addresses to each leaf and spine in the fabric. These IP addresses are contained in a Tunnel Endpoint (TEP) pool which is defined during the initial fabric bring up.

IS-IS is the underlay routing protocol in an ACI fabric. The underlay is automatically build during the fabric bring up and the user does not need to concern themselves with managing this piece of configuration. IS-IS is then responsible for advertising the loopback addresses (TEP IPs) between the leafs and spines.

IS-IS is responsible for infrastructure connectivity in Cisco ACI:

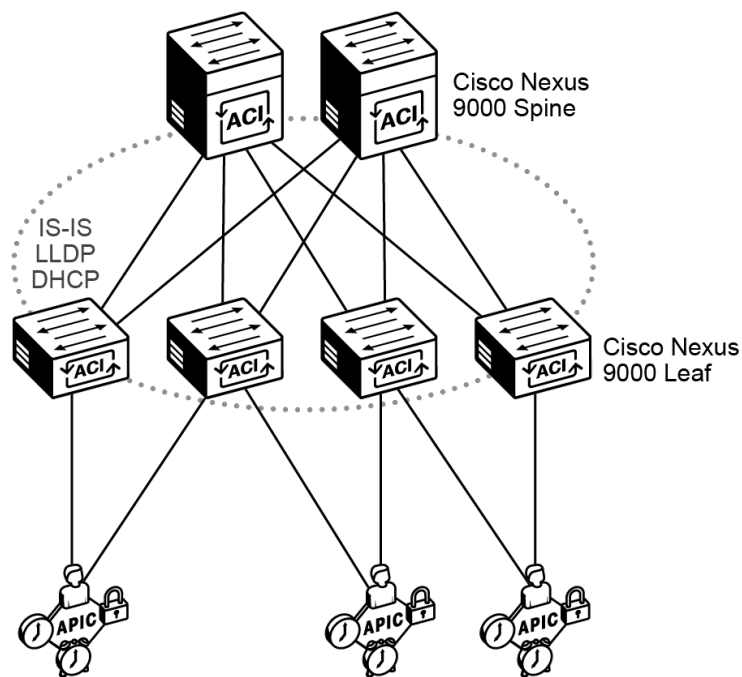
- IS-IS provides IP reachability among TEP addresses.
- It is automatically deployed, and no user intervention is necessary.
- No IS-IS knowledge is required.



Cisco APIC

Cisco APIC is a policy controller. It relays the intended state of the policy to the fabric. The APIC does not represent the control plane and does not sit in the traffic path. The hardware consists of a group of three or more servers in a highly redundant array.

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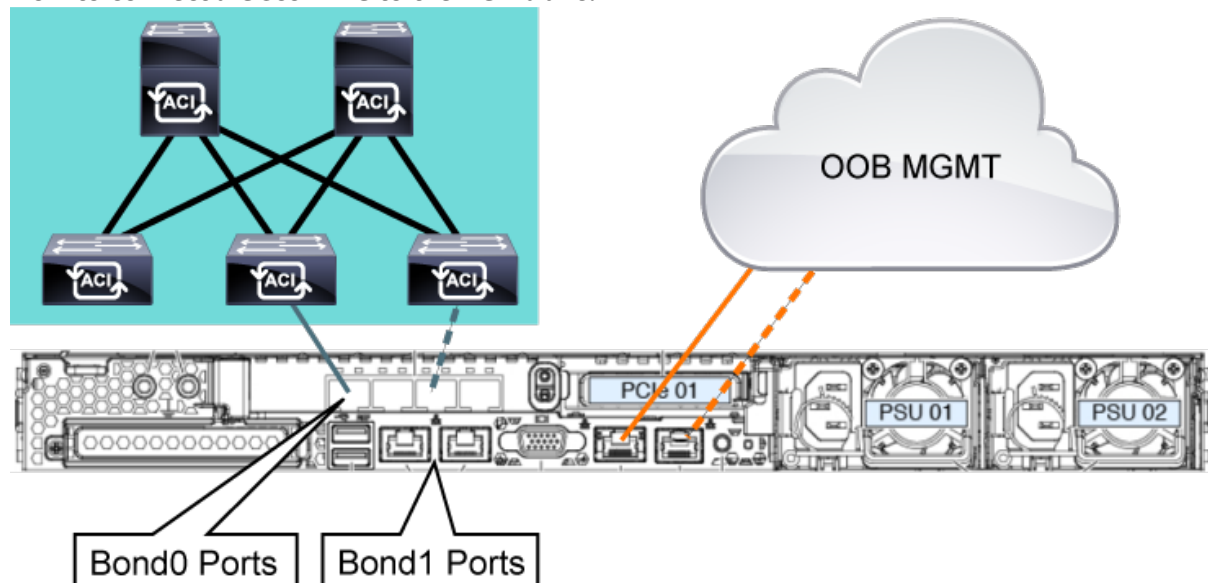


The APIC is a key piece of infrastructure when it comes to the ACI technology. The APIC is the first point of contact when it comes to fabric bring up. LLDP is the layer 2 technology that is used to discover fabric nodes (leafs and spines). DHCP is used to assign a dedicated IP address to each leaf and spine in the fabric. The assigned loopback addresses are the VXLAN TEPs used for IS-IS routing.

The Cisco APIC software is delivered on the Cisco UCS server appliances with a Secure Socket Layer (SSL) certificate that allows the hardware to run Cisco APIC software. Since the hardware is Cisco UCS C-Series, it comes with Cisco Integrated Management Controller (Cisco IMC). The Cisco IMC should run a compatible version from an APIC release note as well.

The Cisco APIC is deployed in a group with a minimum of three controllers, to form a cluster.

How to connect a Cisco APIC to the ACI Fabric?



Cisco APICs use a bonded interface that is typically dual-homed to two leaf switches for connectivity to the Cisco ACI fabric, and have the ability to use a bonded interface that can be dual-homed to the out-of-band management network.

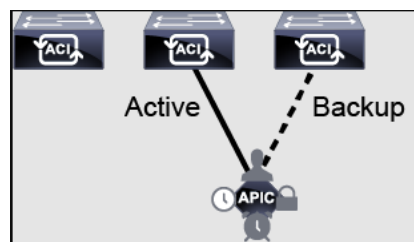
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There are two bond interfaces:

- **Bond0** is used to connect to the fabric itself (to connect to leaf switches that connect into the fabric). It is recommended to connect two fabric uplinks, each to a separate leaf and/or vPC Leaf pairs.
- **Bond1** is used to connect to the out-of-band (OOB) segment (that allows setup of the APIC).

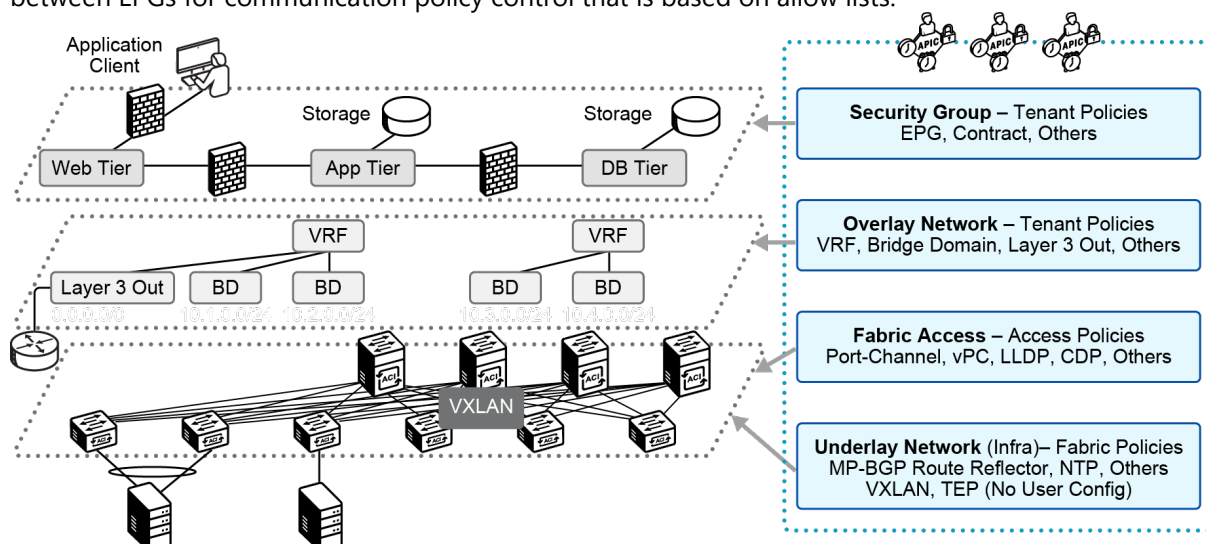
The four ports on the Cisco UCS VIC 1455 are paired in the following manner: port-1 and port-2 (from left to right) represent one pair, corresponding to eth2-1 on APIC and port-3 and port-4 is another pair, corresponding to eth2-2 on APIC.

Eth2-1 and eth2-2 form the Bond0, which is an active/standby Port Channel. Only one connection is allowed for each pair. For example, you can connect one cable to either port-1 or port-2, and connect another cable to either port-3 or port-4 (please do not connect two cables on any pair).



Cisco ACI Policy Model

The policy model manages the entire fabric, including infrastructure, authentication, security, services, applications, and diagnostics. Logical constructs in the policy model define how the fabric meets the needs of any fabric function. The logical construct of a typical Cisco ACI tenant network consists of tenants, VRFs, bridge domains (BDs), endpoint groups (EPGs), and contracts. Contracts are applied between EPGs for communication policy control that is based on allow lists.



The Cisco ACI manages all the configurations as objects stored in the **management information tree (MIT)** database.

As an overview, policies can be categorized in four major groups, as shown in the figure:

- **Security Group (Security and Application Tenant Policies):** Defines how the Cisco ACI fabric should allow network communication between two devices. Example policies are EPGs and contracts, among others.

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- **Overlay Network (Network Tenant Policies):** Defines the overlay network that goes over the Cisco ACI fabric. These policies will be the logical topology of user networks, and you can define them without worrying about the underlay. Example policies include VRF and bridge domain, among others.
- **Fabric Access (Access Policies):** Defines how end hosts or external network devices connect to the Cisco ACI fabric by configuring port-channel, vPC, and other interface-level configurations.
- **Underlay Network (Fabric Policies):** Defines the VXLAN underlay built by the Cisco ACI fabric. It includes the infrastructure Multiprotocol Border Gateway Protocol (MP-BGP) and other protocols to manage Cisco ACI switches such as NTP.

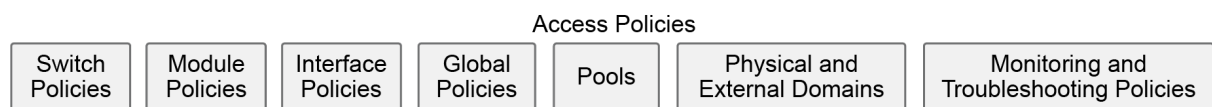
Cisco ACI Access Policies

Access policies configure external-facing interfaces that connect to devices such as hypervisors, hosts, network-attached storage, routers, or fabric extender interfaces.

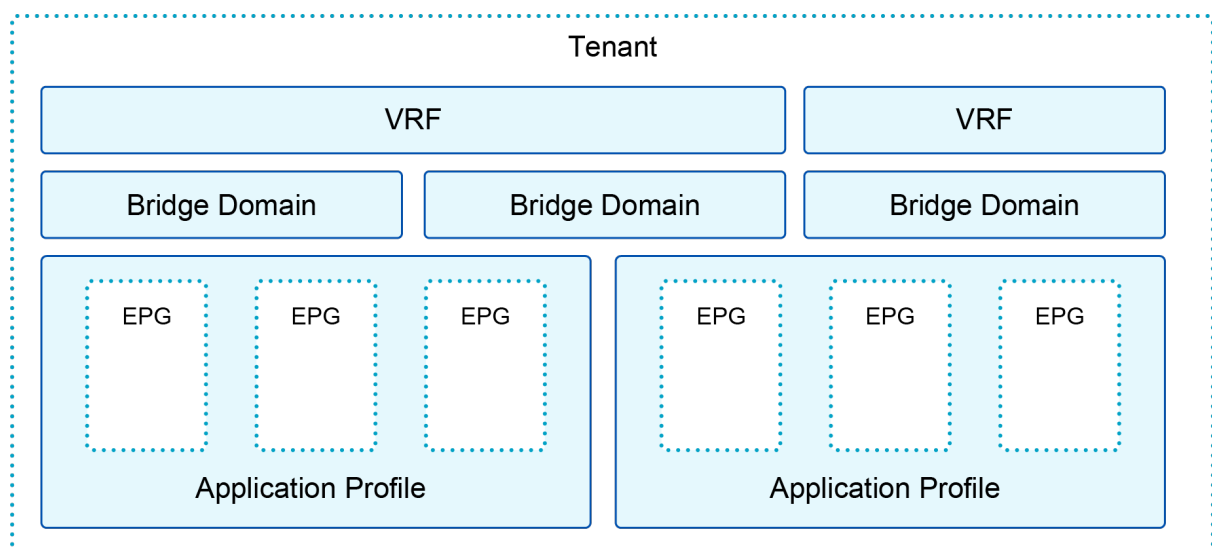
Access policies enable the configuration of port channels and virtual port channels, protocols such as LLDP, Cisco Discovery Protocol, or Link Aggregation Control Protocol (LACP).

Access policies have these functions:

1. Attaching any endpoints to leaf switches (mandatory)
2. Identifying the access interfaces on Cisco ACI switches
3. Authorizing encapsulation resources (VLAN)



Cisco ACI Logical Constructs



Tenant:

- A tenant is a logical container for application policies.
- An ACI fabric can contain multiple tenants

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- A tenant represents a unit of isolation from a policy perspective, but it does not represent a VRF. Tenants can represent a customer in a service provider setting or an organization or domain in an enterprise setting

VRF:

A VRF provides an IP address spaces and a Layer 3 forwarding domain just like a normal router. Each tenant has its own VRFs.

Bridge Domain

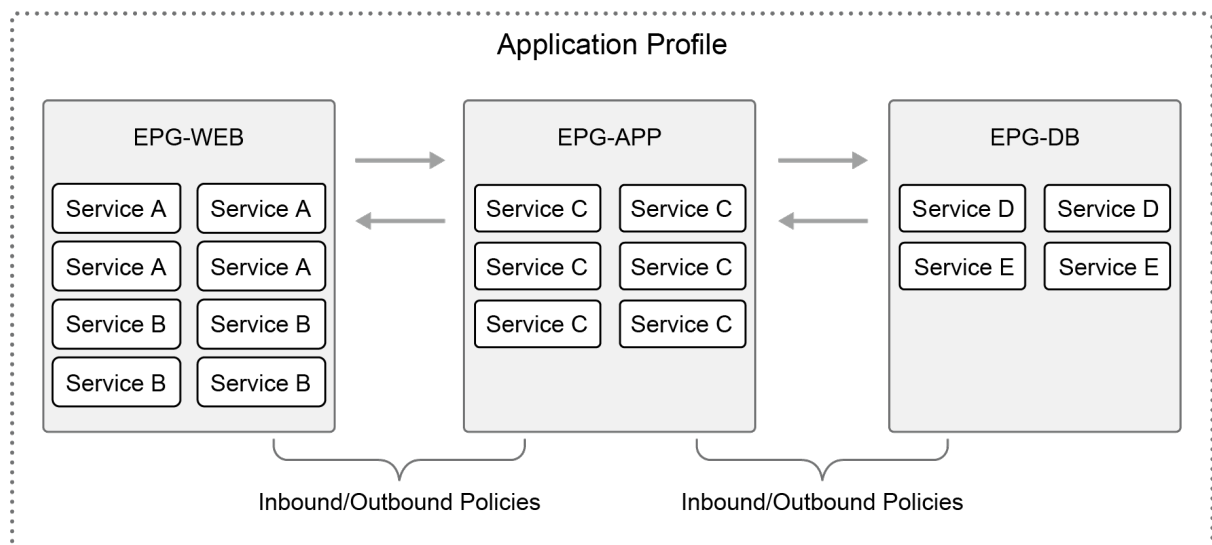
In Cisco ACI, instead of VLAN, a bridge domain is the Layer 2 Forwarding domain within the fabric. When a packet arrives, Cisco ACI maps the VLAN into an EPG. Each EPG belongs to a bridge domain that is the Cisco ACI Layer 2 domain. It performs bridging or MAC address lookup within the bridge domain.

Endpoint Group (EPG)

EPGs to create logical groupings of hosts or servers (endpoints) that perform similar functions within the fabric. An EPG or multiple EPGs can belong to the same Bridge Domain. Since an EPG is for security separation, no endpoints can talk to each other across EPGs unless it is explicitly allowed. Endpoints in the same EPG can talk to each other unless intra-EPG isolation is defined with rules that restrict endpoint communication.

Application Profile

Application profiles are logical containers for grouping EPGs.



Contracts

The inbound/outbound policies that control the communication between a pair of EPGs are known as contracts. Contracts provide a way for the Cisco ACI administrator to control traffic flow within the Cisco ACI fabric between EPGs. You use a provider-consumer model to build contracts, where one EPG provides the services that it wants to offer, and another EPG consumes the services.

Contracts consist of one or more subjects. Each subject contains one or more filters. Each filter contains one or more entries. Each entry is equivalent to an Access Control Entry (ACE) in an access control list (ACL) that applies between EPGs. When the leaf knows the source and destination EPG for a packet, it checks the contract rules between them and decides whether to allow the packet.

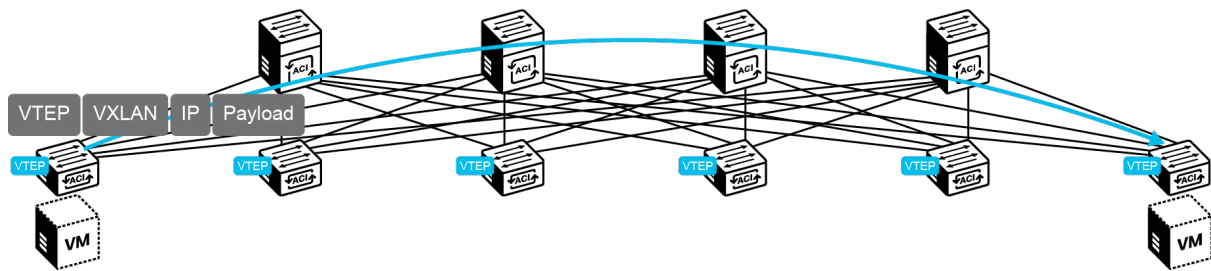
The following are the core principles of contracts:

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- They define the policies using subjects and filters (the type of traffic to allow or deny) between EPGs.
- Communication between EPGs is not allowed without a contract.
- The traffic direction is based on how the contract is attached to EPGs (provider/consumer).

Cisco ACI Endpoint Learning & Traffic Forwarding

Traffic Forwarding



User data traffic is encapsulated with the VXLAN header when being forwarded to another leaf. The forwarding across switch nodes is performed based on the TEP IP in the VXLAN encapsulation.

In case the ingress leaf is unaware of the destination endpoint location (TEP), Cisco ACI has a distributed database that is called the Council of Oracles Protocol (COOP) on each spine. It knows all the mapping of endpoint and TEP.

For more labs visit my GitHub repo: <https://github.com/TitusM/Cisco-Data-Center>

References

Cisco U courses:

1. Understanding Cisco Data Center Foundations | DCFNDU

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